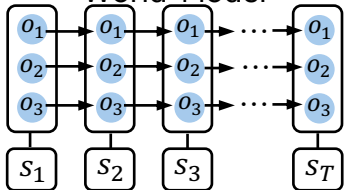


# Wake-Active Inference

## World Model

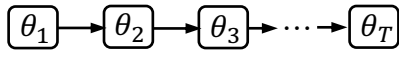


## Reconstruction

$$O_t \sim p(O_t | O_{t-k:t-1}, A_{t-k:t-1})$$

$$S_t \sim p(S_t | O_{t-k:t}, \theta_{t-k:t-1}, A_{t-k:t-1})$$

## Internal Model



## Uncertainty Measurement

$$\theta_t \sim p(\theta_t | S_t, \theta_{t-1}, O_t, z_t)$$

## Policy Model

### Switching

$$a_t \sim \pi(a_t | O_t, S_t, \theta_t)$$

Rollout + Beam Search  $\rightarrow$  Action  $a_t$

### Habitual Engine

$$\hat{a}_t = a_t, \text{ if } I(C_t, O_t) = 1$$

$$R_t = (C_t, a_t), C_t = (P_1(O_t), P_2(O_t), \dots, P_m(O_t))$$

### Rule-driven Reasoning

Rule Pooling

## Goal-Minimize Free Energy

## Sleep-Model Optimization

### Loss of Free Energy

$$F_t^{(i)} = \underbrace{\alpha VFE_t^{(i)}}_{\text{recons}+KL_S} + \underbrace{\beta VFE_t^{(i)}}_{\text{explore.vs.exploit}} + \underbrace{\gamma_t KL_\theta^{(i)}}_{KL_\theta},$$

$$\ell = \ell_{FE} = \frac{1}{B(T-1)} \sum_{i=1}^B \sum_{t=1}^{T-1} F_t^{(i)}$$

### Model Update

$$(\text{belief})\phi \leftarrow \phi - \eta \nabla_\phi \ell, (\text{policy})\varphi \leftarrow \varphi - \eta \nabla \varphi \ell$$

## Sleep-Rule Management

### Rule Candidate Pool

Confidence Level decays

$$\text{conf}_{\text{new}} = \lambda \text{conf}_{\text{old}} + (1 - \lambda) \Delta F$$

### Compound Scoring

$$\text{freq}_a = \frac{\#\{e : e.\text{action} = a\}}{\# \text{events}}, \sup, \overline{\Delta F} = \text{mean}(\dots)$$

Rule Pruning

### Logic Rule Learner

$$\text{freq} < \text{threshold} \rightarrow \text{weight}_a \propto w_f \text{freq}_a + w_s \frac{\sup - \min}{\max - \min} + w_d \frac{\overline{\Delta F} - \min}{\max - \min}$$